

DBMS Sessional Exam — Past-Paper Analysis

Sessional syllabus only | 4 past papers analysed: Jan-2022, Feb-2022 (Supplementary), Jan-2023, Jan-2025 | Nirma University, MCA Sem-I

Key findings at a glance

- **165 of 300 marks (55%)** across the four papers fall inside your sessional content, which matches your estimate of roughly 60%.
- **Five topics make up about 81% of those marks:** CREATE/ALTER TABLE, Normalization, SQL Keys & Referential Integrity, Joins & Set Operations, and DBA roles.
- **Normalization and SQL keys appeared in all 4 papers**, and every paper had at least one DDL question (CREATE or ALTER TABLE) and at least one joins/set-operations question.
- **The same 5-table university schema** (department, course, instructor, section, teaches) was set for CREATE TABLE in 3 of 4 papers. Practise it until you can write it from memory.

How I divided your syllabus into units

Your syllabus PDF (105 pages) has no printed unit headings, so I grouped it into four units by topic. Page numbers refer to your PDF.

Unit	PDF pages	Content
Unit 1 — Introduction, Data Models & Architecture	pp. 1–30	Purpose of DBMS, file-processing disadvantages, three levels of abstraction, data models (relational, ER, object-based, semistructured), DDL/DML, 1/2/3-tier architecture, database users, DBA.
Unit 2 — Relational Model & Keys	pp. 31–46	Relational model, null values, super / candidate / primary / alternate / foreign / composite / artificial keys, integrity constraints, referential integrity.
Unit 3 — SQL	pp. 47–97	Oracle joins (inner, left, right, full outer, cross, natural), parts of SQL, CREATE / ALTER / DROP, basic queries, set operations, nulls, aggregates, GROUP BY / HAVING, nested subqueries, INSERT / UPDATE / DELETE.
Unit 4 — Normalization	pp. 98–105	1NF, 2NF, 3NF and BCNF with worked examples.

1. Topic Weightage Table

Frequency = number of the 4 past papers in which the topic was asked. Marks are for the question as printed; where a question bundles several topics (for example keys + referential integrity), the marks are shared.

Unit	Specific topic	Frequency (of 4 papers)	Where it appeared	Typical marks	Priority
Unit 1 — Introduction, Data Models & Architecture (pp. 1–30)					
Unit 1	Database Administrator (DBA): roles and functions	2/4	Jan-23 Q2A (9); Jan-25 Q1A (8)	8–9	High
Unit 1	Data models and their categories	2/4	Jan-22 Q1A (6); Feb-22 Q1C-OR (6)	6	Medium
Unit 1	File-processing system: pros and cons	1/4	Jan-23 Q1B (6)	6	Medium
Unit 1	DBMS definition and primary goal	1/4	Jan-23 Q1A (6, shared with referential integrity)	6 (shared)	Medium
Unit 1	Components of a database system with diagram (architecture, users)	1/4	Feb-22 Q2A (7)	7	Medium
Unit 1	ER model and mapping cardinality	1/4	Jan-23 Q3B (7)	7	Low–Medium
Unit 2 — Relational Model & Keys (pp. 31–46)					
Unit 2	SQL keys: super, candidate, composite, alternate (plus primary, foreign)	4/4	Jan-22 Q1B (6); Feb-22 Q1C (6); Jan-23 Q1C (6, super and candidate only); Jan-25 Q1B (8)	6–8	Very High
Unit 2	Referential integrity with example	2/4	Jan-23 Q1A (6, shared); Jan-25 Q1B (8, shared)	6–8 (shared)	High
Unit 2	Artificial key	1/4	Jan-25 Q1B (part of 8)	Part of 8	Medium
Unit 3 — SQL (pp. 47–97)					
Unit 3	CREATE TABLE for the university schema (department, course, instructor, section, teaches)	3/4	Jan-22 Q3A (9); Feb-22 Q3B (9); Jan-23 Q2B (9)	9	Very High
Unit 3	ALTER TABLE: add, drop, rename column, modify datatype, add NOT NULL	1/4	Jan-25 Q2B (8)	8	High
Unit 3	Set operations: UNION, INTERSECT, EXCEPT	2/4	Jan-22 Q1C (6); Feb-22 Q1A (6)	6	High
Unit 3	Joins: inner, left, right, full outer, cross	2/4	Jan-23 Q4C (6, with HAVING and EXISTS); Jan-25 Q2A (8, full outer and cross)	6–8	High

Unit	Specific topic	Frequency (of 4 papers)	Where it appeared	Typical marks	Priority
Unit 3	HAVING clause and EXISTS clause	1/4	Jan-23 Q4C (part of 6)	Part of 6	Medium
Unit 4 — Normalization (pp. 98–105)					
Unit 4	Normalization: importance, 1NF, 2NF, 3NF with examples	4/4	Jan-22 Q2B (7); Feb-22 Q2B (7); Jan-23 Q3A (7); Jan-25 Q3A (8)	7–8	Very High

1B. In your syllabus but not yet asked (potential surprises)

These have a 0/4 record in past papers, but they are in your sessional content and are natural counterparts of topics that were asked. The chance column is my own judgement, not a statistic.

Topic	PDF pages	Why it could still appear	Chance of appearing
BCNF (Boyce-Codd Normal Form)	pp. 104–105	Natural extension of the 1NF–3NF question that is asked every year.	Medium
Three levels of abstraction and data independence	pp. 4–6	Classic 5–6 mark theory question; fits with the DBMS definition and file-processing questions.	Medium
Nested subqueries: IN, SOME, ALL, EXISTS	pp. 86–94	EXISTS was asked once; subqueries are the biggest SQL block in your PDF.	Medium
DDL vs DML (database languages)	pp. 12–15	Short compare-and-contrast question with SQL examples.	Medium
1-tier, 2-tier, 3-tier architecture; types of database users	pp. 25–29	Related to the Feb-22 "components of a database system" question.	Medium
Aggregate functions and GROUP BY	pp. 81–86	HAVING was asked with it; practical query questions often use both.	Low–Medium
INSERT, UPDATE, DELETE statements	pp. 95–97	Easy practical SQL; can be bundled into a CREATE/ALTER style question.	Low–Medium
Integrity constraints: domain constraints, assertions	pp. 16–17, 46	Linked to the referential integrity question.	Low–Medium
Natural join, null values	pp. 34, 56–57, 81–86	Minor topics; could appear as part of a joins or aggregates question.	Low

2. High-Yield Priority List

These five topics account for **133 of the 165 sessional-scope marks (about 81%)** in the past papers. Per paper: Jan-22 80%, Feb-22 82%, Jan-23 66%, Jan-25 100%. Ranked by cumulative marks.

Rank	Topic to prioritise	Evidence from past papers	Marks at stake	What to prepare
1	DDL: CREATE TABLE and ALTER TABLE	A DDL question appeared in all 4 papers : CREATE TABLE for the university schema in Jan-22, Feb-22 and Jan-23 (9 marks each), ALTER TABLE in Jan-25 (8). Highest marks per question of any sessional topic.	8–9 per question; 35 marks total (26%)	- Write the 5-table script from memory: varchar/numeric types, PKs (composite for section and teaches), FKs, NOT NULL. - Know 5 ALTER TABLE forms: ADD, DROP COLUMN, RENAME COLUMN, MODIFY datatype, add NOT NULL. - State your assumptions (data types, keys) in a line, because the paper asks for it.
2	Normalization (1NF → 3NF)	4/4 papers with near-identical wording: why normalization is important, then 1NF, 2NF and 3NF with examples.	7–8 per question; 29 marks total (22%)	- Define normalization and its goal (less redundancy; no insert, update or delete anomalies). - Take ONE example table through 1NF → 2NF → 3NF with before/after tables (your PDF pp. 98–105 has teacher and emp_zip examples). - One-line rules: atomic values / no partial dependency / no transitive dependency. - Add BCNF in 2–3 lines as a bonus (pp. 104–105).
3	SQL Keys and Referential Integrity	Keys in 4/4 papers (super, candidate, composite, alternate; artificial in Jan-25). Referential integrity in Jan-23 and Jan-25. Keys are usually opened with "Why do we need..." (3 of 4 papers).	6–8 per question; 26 marks total (20%)	- Use one STUDENT table (pp. 35–43) to show every key type. - Remember: every primary key is a candidate key, and every candidate key is a super key. - Start with why keys are needed: unique identification, linking tables, avoiding duplicates. - Referential integrity: parent/child tables, foreign key, what happens on an invalid insert or delete. - Artificial key: used when the natural key is large or complex (p. 44).
4	Joins and Set Operations	At least one per paper: UNION/INTERSECT/EXCEPT in both 2022 papers; joins in Jan-23 (inner, left, right, full outer, plus HAVING and EXISTS) and Jan-25 (full outer and cross join, using employees and departments).	6–8 per question; 26 marks total (20%)	- For each join: definition + Venn diagram + query + sample output on two small tables (inner, left, right, full outer, cross; natural as an extra). - UNION vs UNION ALL, INTERSECT, EXCEPT with the course/semester examples on pp. 78–80; mention that Oracle uses MINUS. - One worked query each for HAVING (with GROUP BY) and EXISTS (pp. 84–94).

Rank	Topic to prioritise	Evidence from past papers	Marks at stake	What to prepare
5	DBA: roles and functions	2/4 papers , and both 100-mark papers asked it: Jan-23 (9 marks) and Jan-25 (8 marks).	8–9 per question; 17 marks total (13%)	- Learn the five functions on pp. 29–30: schema definition; storage structure and access-method definition; schema and physical-organization modification; granting authorization; routine maintenance (backup, disk space, monitoring). - Write 2 lines per function to reach 8–9 marks.

Percentages are shares of the 133 marks covered by these five topics and are rounded.

Next in line (if you still have time after the top five)

Topic	Frequency	Why it is worth it
Data models and their categories	2/4	Asked in both 2022 papers (6 marks). Short, definition-driven, cheap marks (pp. 8–12).
File-processing system pros/cons; DBMS definition and goal	1/4	Both in Jan-23 (6 marks each). Your PDF lists seven disadvantages (pp. 3–4), which are easy to memorise.
Database system components / architecture with diagram	1/4	Feb-22 (7 marks) asked for a diagram. Cover 1/2/3-tier architecture, users and the DBA (pp. 25–30).
ER model and mapping cardinality	1/4	Jan-23 (7 marks). Your PDF only introduces mapping cardinality (pp. 9–10, 22–23), so add one-to-one, one-to-many, many-to-one and many-to-many from your class notes, with an ER diagram.

3. Question Patterns

Pattern	What it looks like	Evidence	How to use it
1. Questions repeat almost verbatim	The same question text returns in later papers with only small changes.	Normalization 4/4; keys 4/4; university-schema CREATE TABLE 3/4; set operations in both 2022 papers; DBA and referential integrity in both recent papers.	Treat past papers as your question bank. Practise each recurring question as a full written answer.
2. Fixed-schema DDL question	"Write SQL for creating a table corresponding to the schema..." with department, course, instructor, section, teaches, plus "make any reasonable assumptions about data types, primary key and reference key".	Jan-22 Q3A, Feb-22 Q3B, Jan-23 Q2B: identical schema, 9 marks each.	Memorise the schema and its keys (composite PKs on section and teaches; FKs to department, course, section, instructor).
3. Bundle questions	One 6–8 mark question packs 3–6 concepts: "super, candidate, composite, alternative key"; "union, intersect, except"; "inner, left, right, full outer, having, exists"; "referential integrity + alternate + artificial keys".	Every paper has at least one bundle question.	Give each concept a crisp definition and one example. Budget about 1–1.5 marks per concept; do not over-write one item.

Pattern	What it looks like	Evidence	How to use it
4. "Why is X important / needed?" opener	The question starts with motivation before the list of concepts.	5 questions in 3 papers: "Why do we need a SQL key" (Jan-22, Feb-22), "Why SQL key is important" (Jan-23), "Why normalization ... is important" (Jan-22, Feb-22).	Open your answer with a 2–3 line importance statement, then move to definitions.
5. "With appropriate example(s)" and "assume and mention data"	Examples are explicitly demanded, and every paper instructs you to assume suitable data and mention it.	14 of the 23 in-scope questions say "with (an) appropriate example(s)"; the "assume suitable data" instruction is on all 4 papers.	Never write definitions alone. Always include a small sample table and say "Assumed data:" before it.
6. Diagram-based answers	"...with an appropriate diagram"; instructions say "Draw neat sketches wherever necessary"; ER diagrams for cardinality.	Feb-22 Q2A (7 marks, diagram); Jan-23 Q3B (ER); all 4 papers carry the sketch instruction.	Practise three diagrams: database system structure / 3-tier, an ER diagram with cardinality, and Venn diagrams for joins.
7. Sessional content sits in the front of the paper	Sessional-type topics fill Q1–Q2 (and part of Q3); PL/SQL, transactions and indexing sit in the later questions and Section II.	Jan-23 Section I: 50 of 50 marks in scope. Jan-25 Section I: 40 of 50. Jan-25 Section II: 0 of 50.	Expect your sessional to look like the front half of these papers: theory in Q1–Q2, then a SQL/DDI question.
8. Questions are getting more applied	2022: broad "discuss" prompts. 2025: concrete tasks on named tables (employees, departments) and a five-part ALTER TABLE; every part is 8 marks.	Jan-25 Q2A (employees and departments), Q2B (five ALTER operations).	Practise writing queries and outputs on small named tables, not just theory.

3B. Difficulty tags (Bloom's level) by question family

Question family	Bloom's levels seen	What is actually tested
SQL keys	BL-1 (Jan-22, Feb-22); BL-2 (Jan-23); BL-3 (Jan-25)	Recall definitions, then apply them to an example table.
Normalization	BL-1 (Jan-22); BL-3 (Jan-23, Jan-25); BL-4 (Feb-22)	Explain each form and apply it to a worked example.
CREATE TABLE schema	BL-6 (Jan-22, Feb-22); BL-4 (Jan-23)	Create: write correct DDL with keys and constraints.
Set operations	BL-4 (Jan-22, Feb-22)	Analyse: write queries and explain results.
Joins / HAVING / EXISTS	BL-5 (Jan-23); BL-2 (Jan-25)	Write queries; explain how each join works.
ALTER TABLE	BL-4 (Jan-25)	Correct syntax plus an example for each of five operations.
DBA; data models	DBA: BL-3 (Jan-23), BL-2 (Jan-25); data models: BL-2 (Jan-22, Feb-22)	Understand and describe in detail.

The same question carries different BL tags in different years, so judge depth by the marks (6–9), not by the tag.

Appendix A. Scope check: what I counted as sessional

Paper	Max marks	In-scope marks	In-scope %	Treated as outside the sessional syllabus
Jan-2022 (3MCA104, Semester End)	50	34	68%	User-defined exception (Q2A); trigger (Q3B) and explicit cursor (Q3B-OR); ACID (Q1C-OR)
Feb-2022 (3MCA104, Supplementary)	50	35	70%	Locking protocols (Q1B); trigger (Q3A); implicit cursor (Q3B-OR)
Jan-2023 (6CS104)	100	56	56%	Section II: indexing (Q4A), views (Q4B), implicit cursor (Q5A), trigger (Q5B), exception (Q6A), ACID (Q6B)
Jan-2025 (6CS104CC22)	100	40	40%	Explicit cursor (Q3B); all of Section II: locking, timestamp ordering, indexing (2 questions), exception, trigger
Total	300	165	55%	Where a question had an OR choice, I counted the in-scope option once.

Topics left out because they are not in your sessional PDF

Topic group	Appeared in	Note
PL/SQL: triggers	4/4 papers (Feb-22, Jan-22, Jan-23, Jan-25)	Your PDF mentions PL/SQL only as an example of a procedural DML (p. 13).
PL/SQL: user-defined exceptions	3/4 papers (Jan-22, Jan-23, Jan-25)	Not in the sessional PDF.
PL/SQL: implicit and explicit cursors	Implicit: Feb-22, Jan-23. Explicit: Jan-22, Jan-25	Not in the sessional PDF.
Transactions and ACID	Jan-22 (OR), Jan-23	The PDF lists COMMIT/ROLLBACK in one table (p. 59) but does not cover ACID.
Locking protocols; timestamp ordering	Locking: Feb-22, Jan-25. Timestamp: Jan-25	Concurrency control is not in the PDF.
Indexing: hashing, multilevel, sparse, dense, clustering, secondary	Jan-23 (Q4A); Jan-25 (Q5A, Q5B)	The PDF has only a one-paragraph note on indices (p. 61).
SQL views: create, update, drop	Jan-23 (Q4B)	The PDF covers the view level of abstraction but no CREATE VIEW syntax.

Judgement calls

- **"Components of a database system" (Feb-22 Q2A)** is mapped to architecture, users and the DBA (pp. 25–30).
- **ER model and cardinality (Jan-23 Q3B)** is counted as in scope, but your PDF covers it only briefly (pp. 9–10, 22–23).
- **If your faculty taught any excluded topic before the sessional** (for example views or transactions), treat it as in scope and revise those questions too.

Appendix B. Paper-by-paper map of in-scope questions

Paper	In-scope questions counted (marks in brackets)
Jan-2022 (50 marks)	Q1A Data models and categories (6) Q1B SQL keys (6) Q1C Union / intersect / except (6) Q2B Normalization 1NF–3NF (7) Q3A CREATE TABLE, university schema (9) Total: 34
Feb-2022 (50 marks)	Q1A Union / intersect / except (6) Q1C SQL keys (6) [OR: data models] Q2A Components of a database system with diagram (7) Q2B Normalization 1NF–3NF (7) Q3B CREATE TABLE, university schema (9) Total: 35
Jan-2023 (100 marks)	Q1A DBMS definition, primary goal, referential integrity (6) Q1B File-processing system pros and cons (6) Q1C SQL keys: super and candidate (6) Q2A DBA roles and functions (9) Q2B CREATE TABLE, university schema (9) Q3A Normalization 1NF–3NF (7) Q3B ER model and cardinality (7) Q4C Inner / left / right / full outer join, HAVING, EXISTS (6) Total: 56
Jan-2025 (100 marks)	Q1A DBA roles and responsibilities (8) Q1B Referential integrity, alternate keys, artificial keys (8) Q2A Full outer join and cross join (8) Q2B ALTER TABLE: add, drop, rename, modify, NOT NULL (8) Q3A Normalization 1NF–3NF (8) Total: 40